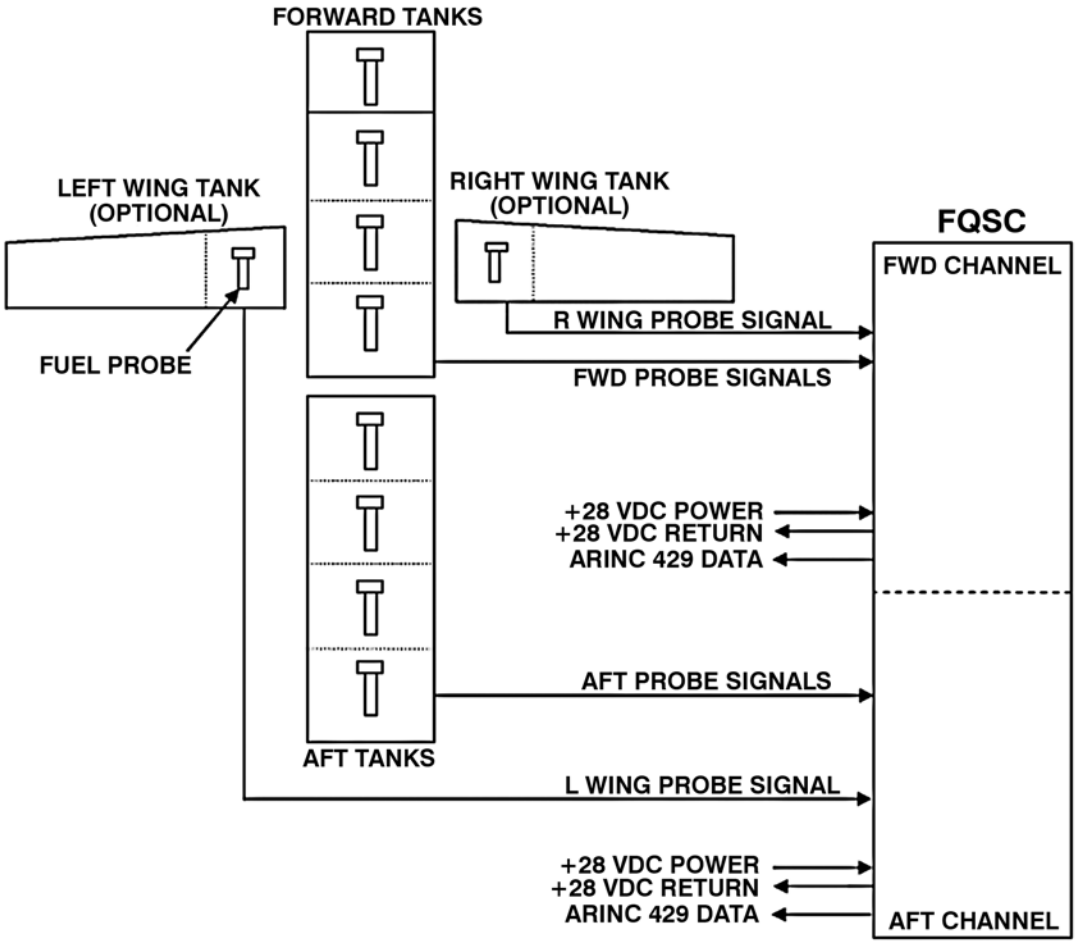
Fuel Quantity Indicating System – General Data

1. General

The Fuel Quantity Gauging System consists of 8 fuel probes in the forward and aft fuel cells, 2 optional fuel probes in wing tanks, and one Fuel Quantity Signal Conditioner (FQSC). The FQSC is a two-channel microprocessor-based unit that interfaces with all the fuel probes on the aircraft and computes fuel quantity. The signal conditioner transmits fuel quantity, fault status and diagnostic information to the cockpit display system via independent forward and aft ARINC–429 data buses. A system level block diagram is provided in Figure 1.



Fuel Quantity System
Figure 1

2. Description

The Fuel Quantity Gauging System consists of a Fuel Quantity Signal Conditioner (FQSC), Main Fuel Probes (Qty 8) and Auxiliary Wing Tank Fuel Probes (Qty 2), if installed.

A. Fuel Quantity Signal Conditioner (FQSC)

The FQSC prevents any electrical ignition sources from entering the fuel tank through the fuel system wiring. One FQSC is installed in the system. The FQSC contains protective components which prevent hazardous levels of voltage, power and stored energy from reaching the in-tank wiring and fuel probes. It also contains electrical interface circuits appropriate to produce a signal to an ARINC–429 bus.

B. Fuel Probes

The fuel probes are fluid level sensors with built-in signal conditioning electronics. The probe detects fluid quantity as a change in capacitance between two concentric tubes. A capacitance measurement circuit produces a DC current output proportional to fuel height. The active fuel probe integrates advanced signal processing electronics into the capacitance sensor. The active probe electronics serve to do an accurate measurement of the sensor capacitance and output a robust DC current signal. This signal is protected against electromagnetic interference (EMI) and lightning induced transients. The outer shell of the probe is bonded to airframe ground, making it self-shielding against electromagnetic interference. Each probe consists of two concentric tubes that form the capacitance-sensing element. At one end of the probe is a sealed housing for the electronic circuits. Electrical connections are made via a military spec circular connector. The FQSC acts as the processing unit.

The main probe's electrical connector (Pre Mod 6/2395) consists of a D38999/20WA35PN (mating D38999/26WA35SN). Post Mod 6/2390 have M85049/38S9W backshell with shield termination. The connector backshells are environmentally sealed with a dual wrap kit consisting of polyurethane tape and backfilled with two part polyurethane sealant to reduce moisture ingress.

For Post Mod 6/2395, the main probe's electrical connector consists of a stainless steel connector D38999/21 YA35PN (mating 233–205–G6–Z1–9–35–SN). Post Mod 6/2395 have 620H084BM09A backshell with shield termination. The connector backshells are backfilled with two part polyurethane (self leveling green) sealant to reduce moisture ingress.

The wing probe's electrical connector consists of a MS3449H8C33P per MIL–C–26482 Series 2 (mating connector MS3476L8–33S).

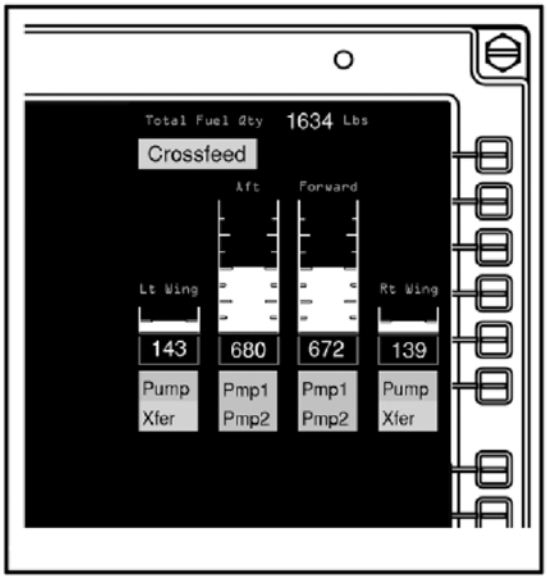
To maintain safety, the current and voltage available via in-tank wiring is limited by components in the FQSC. The probe's signal output circuits incorporate suitable protection against electrical transients and electromagnetic interference.

C. Fuel Quantity

Fuel quantity is displayed on the fuel pane window on Apex. Fuel quantity is provided for each of the aft and forward fuselage tanks, refer to Figure 2. Total fuel quantity of all fuel tanks is also displayed.

D. Wing Tip Tanks Fuel Quantity (Mod S.O.O 6247)

Fuel quantity for the wing tanks is displayed on the fuel pane window on Apex. Fuel quantity is provided for each of the left and right wing tip fuel tanks, refer to Figure 2.



Fuel Indication Display
Figure 2

3. Adjustment/Test

There are no field settable calibration features in the FQSC or the fuel tank probes. The following test is for verification that the resistors in the FQSC are functional and therefore providing the proper level of current limiting to the probes.

A. FQSC Probe Signal Inputs – Current Limit

Temporarily connect an ammeter between each of the FQSC probe signal inputs and the respective return pin, one at a time. Refer to Figure 3 for pin connections. Measure and record the short circuit current for each probe signal input. Current shall be between 5 and 7 mA. Record the findings on tables similar to Table 1 — Fwd Channel (J1) and Table 2 — Aft Channel (J2).

Note Probe 5 for each channel is the associated wing tank probe as per Figure 3. It is not necessary to check this channel on aircraft not equipped with wing tanks.

B. FQSC Probe Signal Input – Short Circuit Current

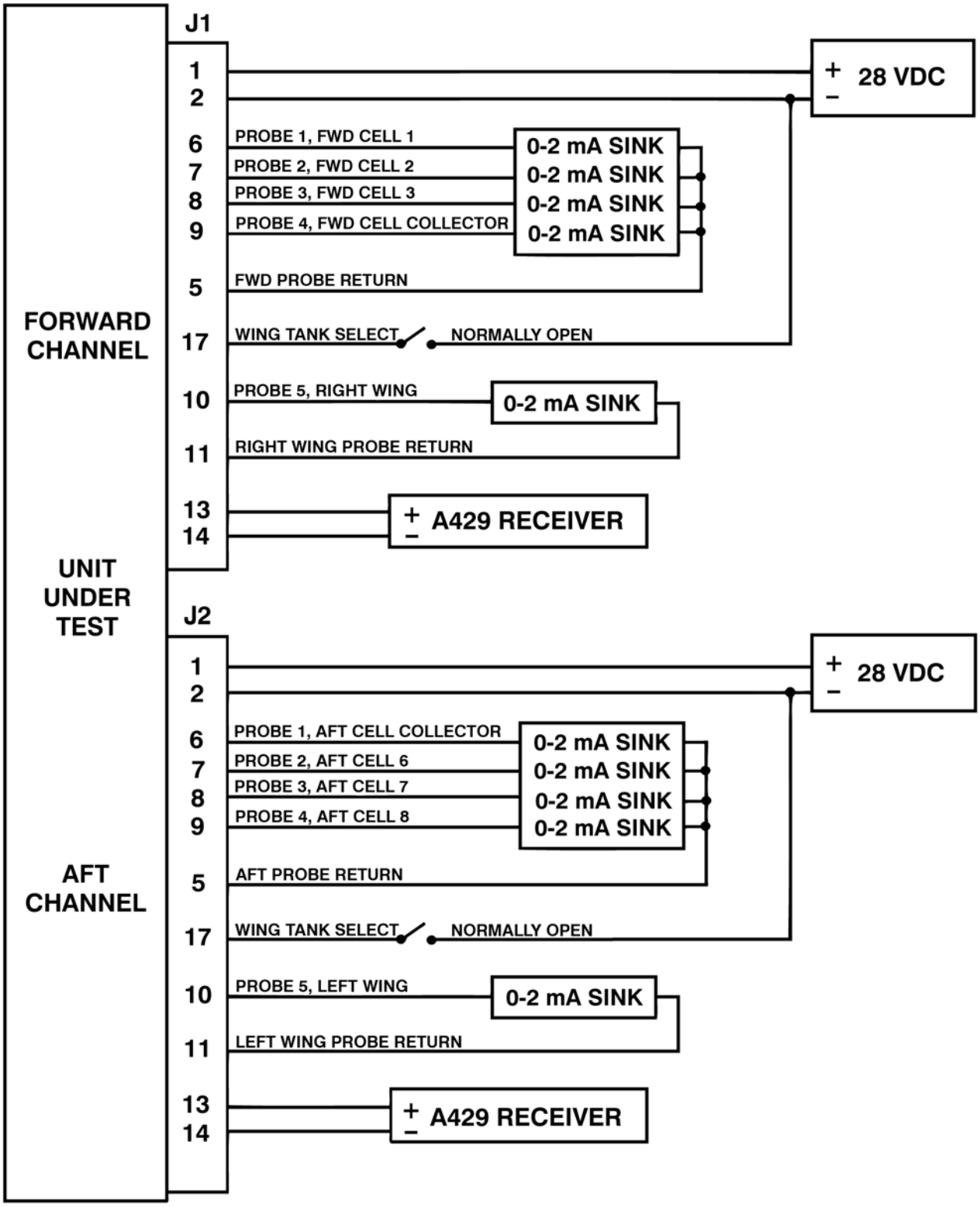
Short circuit current 5.0 < current < 7.0 mA.

Table 1 Fwd Channel (J1)

Probe 1	mA
Probe 2	mA
Probe 3	mA
Probe 4	mA
Probe 5	mA

Table 2 Aft Channel (J2)

Probe 1	mA
Probe 2	mA
Probe 3	mA
Probe 4	mA
Probe 5	mA



Test Connections
Figure 3

C. Calibration Check – Fuselage Tanks (Fuel Tanks Empty)

Note Aircraft must be reasonably level, and in the open air, during all fuel system checks. Wing tanks and components are checked separately to permit testing of fuselage tank system if wing tanks are not to be used.

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Drain all usable fuel from the fuselage tanks and ensure all relevant circuit breakers engaged.

3) On the fuel control panel, verify FWD and AFT BOOST PUMP 1 switches are in the OFF position and FWD and AFT BOOST PUMP 2 switches are in the NORMAL position.

4) Fill the fuselage tanks, forward tank first in increments as listed in Table 3 — Fuel Quantity System Calibration.

5) Check indication readings are within limits shown in Table 3 — Fuel Quantity System Calibration.

6) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

Table 3 Fuel Quantity System Calibration

FUEL IN TANK (PER TENDER METER)	FUEL QUANTITY INDICATION
0	0 – 15 lbs
30 Imp (36 U.S.) gal	215 – 250 lbs
50 Imp (60 U.S.) gal	370 – 410 lbs
100 Imp (120 U.S.) gal	740 – 820 lbs
150 Imp (180 U.S.) gal	1110 – 1230 lbs
Full	1215 – 1240 lbs – Fwd 1320 – 1345 lbs – Aft

Note For usable capacities and fuel grades refer to Table 4 — Fuel Tank Capacities and Fuel Grades. Table 3 — Fuel Quantity System Calibration is based on Jet A–1 fuel density.

Table 4 Fuel Tank Capacities and Fuel Grades

FUEL	DENSITY (AVER-AGE) LB. PER	FORWARD TANK			AFT TANK		
GRADE	IMP. GAL	Weight-lb	Imp. Gal.	U.S. Gal.	Weight-lb	Imp. Gal.	U.S. Gal.
Jet A	8.18	1235.2	151.0	181.2	1341.5	164.0	197.8
Jet A–1	8.16	1232.2	151.0	181.2	1338.2	164.0	197.8
Jet B	7.80	1177.8	151.0	181.2	1279.2	164.0	197.8
JP–4	7.80	1177.8	151.0	181.2	1279.2	164.0	197.8
JP–5	8.16	1232.2	151.0	181.2	1338.2	164.0	197.8
JP–8	8.07	1218.6	151.0	181.2	1325.5	164.0	197.8

D. Calibration Check – Wing Tanks (Fuel Tanks Empty) (Mod S.O.O. 6247)

Note Aircraft must be reasonably level, and in the open air, during all fuel system checks.

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Drain all usable fuel from the wing tanks and ensure all relevant circuit breakers are engaged.

3) On the fuel control panel, verify FWD and AFT BOOST PUMP 1 switches are in the OFF position and FWD and AFT BOOST PUMP 2 switches are in the NORMAL position.

4) Verify the WING TANK fuel switches are in the OFF position.

5) Fill each wing tank in increments as listed in Table 5 — Wing Tanks – Fuel Quantity System Calibration.

6) Check indication readings are within limits shown in Table 5 — Wing Tanks – Fuel Quantity System Calibration.

7) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

Table 5 Wing Tanks – Fuel Quantity System Calibration

FUEL IN TANK (PER TENDER METER)	FUEL QUANTITY INDICATION
0	0 – 10 lbs
11 Imp (13 U.S.) gal	80 – 100 lbs
22 Imp (26 U.S.) gal	165 – 185 lbs
32 Imp (38 U.S.) gal	245 – 265 lbs
Full	285 – 305 lbs

Note For usable capacities and fuel grades refer to Table 4 — Fuel Tank Capacities and Fuel Grades. Table 5 — Wing Tanks – Fuel Quantity System Calibration is based on Jet A–1 fuel density.

4. Maintenance

A. Fuel Quantity Signal Conditioner (FQSC)

Maintenance of the FQSC is ‘on condition' only. On condition maintenance is described as follows: there are no periodic service requirements necessary to maintain continued airworthiness. No maintenance is required until the equipment does not properly do its intended function. If the FQSC is removed or replaced, verify current limiting functionality per procedures of Para 3.A and Para 3.B.

B. Fuel Probes

Maintenance of the main tank fuel probes is ‘on condition' only. On condition maintenance is described as follows: there are no periodic service requirements necessary to maintain continued airworthiness. No maintenance is required until the equipment does not properly do its intended function. For removal and installation of the fuel probes refer to Para 5.If the main tank fuel probes are removed or replaced, do calibration check per procedure of Para 3.C.

C. Wing Tip Tanks Fuel Probes (Mod S.O.O. 6247)

Maintenance of the wing tip fuel tank fuel probes is ‘on condition' only. On condition maintenance is described as follows: there are no periodic service requirements necessary to maintain continued airworthiness. No maintenance is required until the equipment does not properly do its intended function. If the wing tank fuel probes are removed or replaced, do calibration check per procedure of Para 3.D.

5. Removal/Installation

A. Removal of Fuel Probe

1) Ensure POWER SOURCE switch is OFF (Pre Mod 6/2314); or BATTERY and EXTERNAL switches are OFF (Post Mod 6/2314), crossfeed valve is closed, and wing fuel tank switches (if fitted) are OFF.

Note Use appropriate wire stripping and crimping tools for the wire size and contact removal/installation tool.

2) Pull CB151 (FUEL SHUTOFF L), CB153 (FUEL QTY L/AFT), CB 150 (FUEL SHUTOFF R), and CB152 (FUEL QTY R/FWD) circuit breakers on the main circuit breaker panel.

3) Drain fuel from forward tank if No. 1, 2, 3 and/or 4 fuel probes are to be removed. Drain fuel from aft tanks 5, 6, 7 and 8, if fuel probes are to be removed. Drain both tanks if No. 5 and/or 6 cell fuel probes are to be removed. (Refer to 12-10-10, Fuel System – General Data).

4) Gain access to fuel cells under aircraft belly by removing appropriate access panel from underside of fuselage. (Refer to 12-00-00, Servicing – General Data).

5) Disconnect required connectors P113 to P120 from fuel probe, as required.

6) Remove six (6) screws from the fuel probe to the fuel cell sump plate and remove gasket and the fuel probe.

B. Installation of Fuel Probe

1) Visually inspect the environmental sealing kit, if installed on the fuel probe connector. If damaged, replace the environmental sealing kit as per 20-50-03, Environmental Sealing (Dual Wrap Kit) – General Data.

2) Install fuel probe, gasket and secure with screws (6). Safety the screws.

3) Reconnect connectors P113 to P120 to main fuel probe(s).

4) Carry out bonding/resistance measurement between mounting screws and the closet airframe ground or airframe structure to ensure bonding resistance between two points ≤ 2.5 milliohms.

5) Reset CB151 (FUEL SHUTOFF L), CB153 (FUEL QTY L/AFT), CB150 (FUEL SHUTOFF R), and CB152 (FUEL QTY R/FWD) circuit breakers on the main circuit breaker panel.

6) Carry out adjustment test procedures as per Para 3.A to Para 3.C.

7) Re-Install the appropriate access panels from underside of fuselage.